EXPERIMENT NO: 20

Experiment to understand K-means clustering algorithm for a given data set.

PROGRAM:

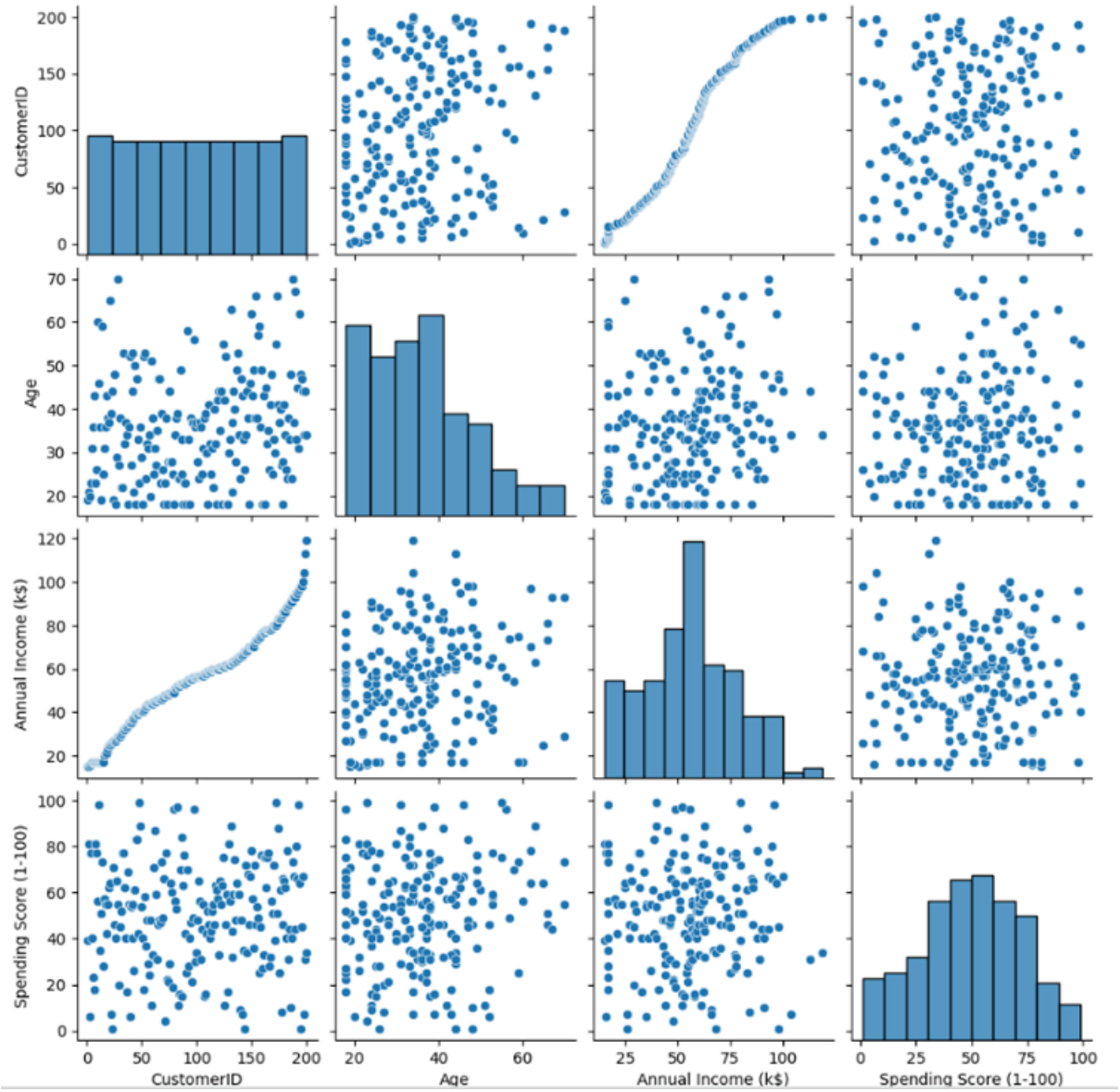
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
df=pd.read_csv('Mall_Customers.csv')
print(df.info())
print(df.head())
sns.pairplot(df)
features=df.iloc[:,[3,4]].values
from sklearn.cluster import KMeans
model=KMeans(n_clusters=5)
model.fit(features)
KMeans(n_clusters=5)
Final=df.iloc[:,[3,4]]
Final['label']=model.predict(features)
print(Final.head())
sns.set_style("whitegrid")
sns.FacetGrid(Final,hue="label",height=8).map(plt.scatter,"Annual Income (k$)","Spending Score (1-100)").add_legend();
plt.show()
features_el=df.iloc[:,[2,3,4]].values
from sklearn.cluster import KMeans
wcss=[]
for i in range(1,10):
```

```
  model=KMeans(n_clusters=i)
  model.fit(features_el)
  wcss.append(model.inertia_)
plt.plot(range(1,10),wcss)
```

OUTPUT:

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 200 entries, 0 to 199
Data columns (total 5 columns):
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
 0   CustomerID              200 non-null    int64
 1   Gender                  200 non-null    object
 2   Age                     200 non-null    int64
 3   Annual Income (k$)      200 non-null    int64
 4   Spending Score (1-100)  200 non-null    int64
dtypes: int64(4), object(1)
memory usage: 7.9+ KB
```

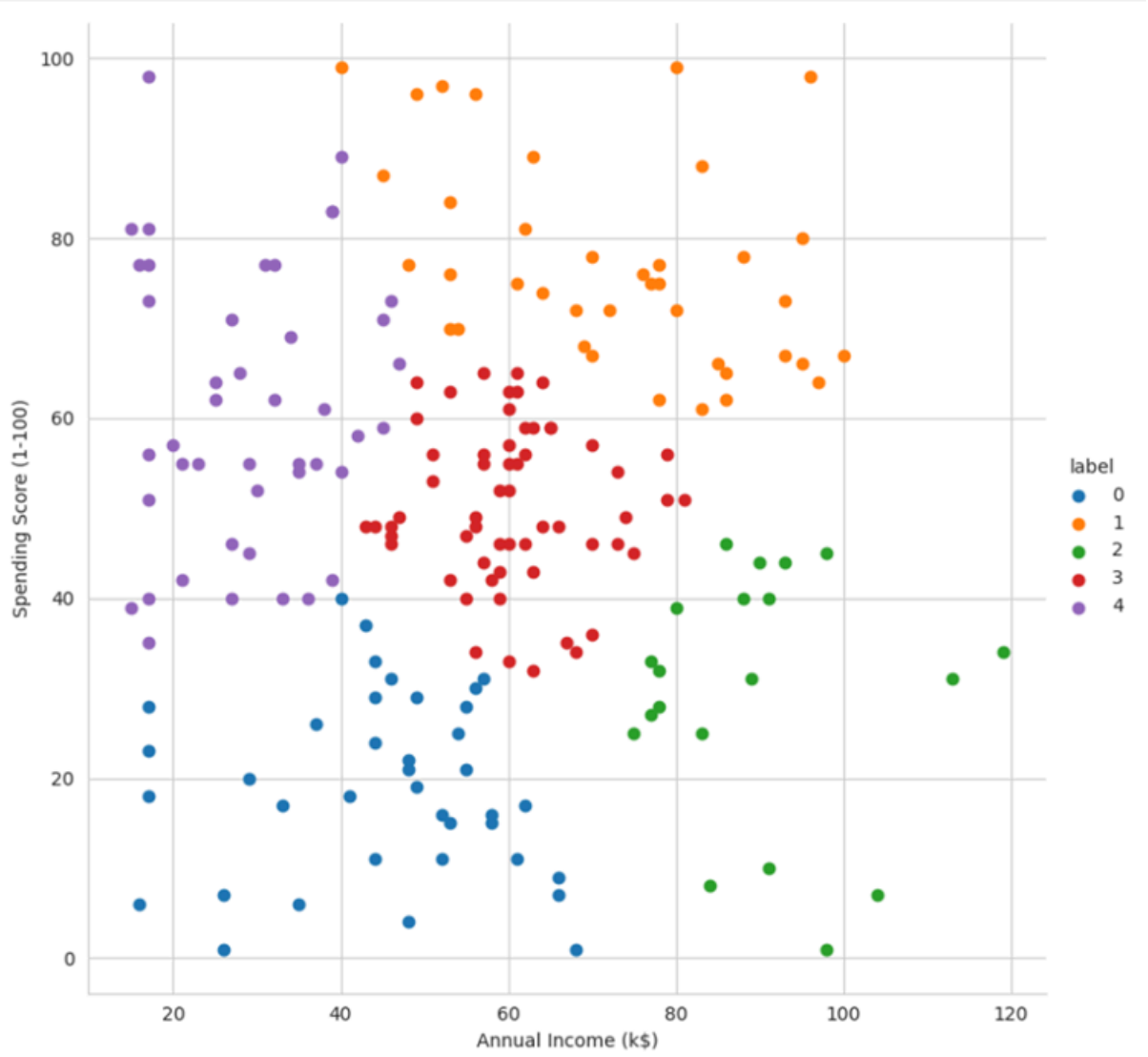
	CustomerID	Gender	Age	Annual Income (k$)	Spending Score (1-100)
0	1	Male	19	15	39
1	2	Male	21	15	81
2	3	Female	20	16	6
3	4	Female	23	16	77
4	5	Female	31	17	40

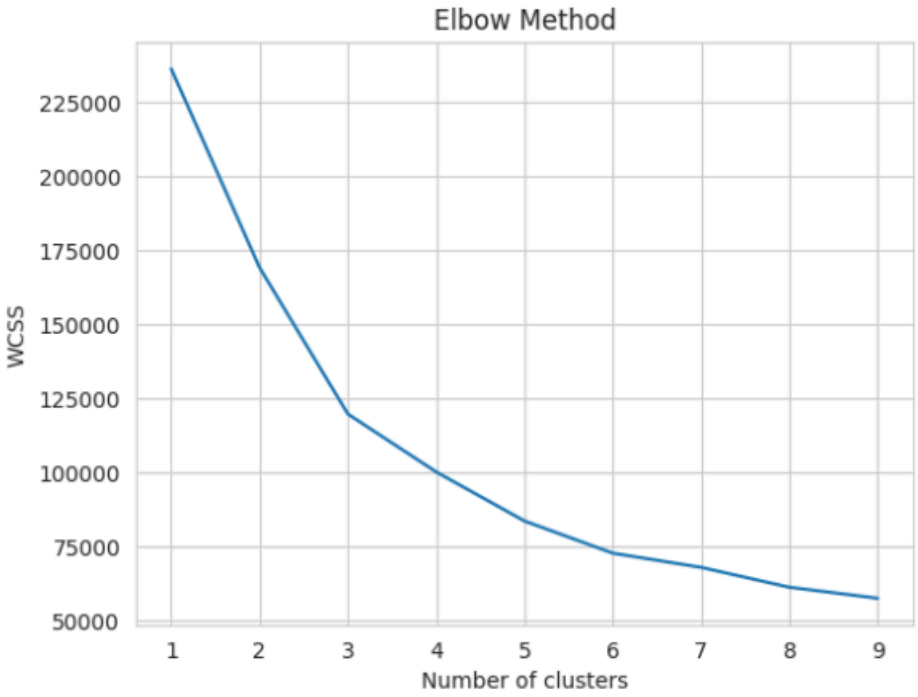


KMeans

KMeans(n_clusters=5)

	Annual Income (k$)	Spending Score (1-100)	label
0	15	39	4
1	15	81	4
2	16	6	0
3	16	77	4
4	17	40	4


Spending Score (1-100)
Annual Income (k$)
label
0
1
2
3
4


Elbow Method
WCSS
225000
200000
175000
150000
125000
100000
75000
50000
1
2
3
4
5
6
7
8
9
Number of clusters